FDRFlow — Technical Architecture & Implementation Report

AICTE Security Deposit / Fixed Deposit Receipt (FDR) Lifecycle Automation Portal

Project Parameter	Specification Details
Problem Statement ID	PS 26265 (Smart India Hackathon 2026)
Target Organization	All India Council for Technical Education (AICTE), Ministry of Education
Category & Domain	Smart Automation / Software Solution
Prototype Status	Fully Functional, Live Local Deployment with Complete E2E Integration
Frontend Environment	http://localhost:5173 (React 18 + Vite SPA with Glassmorphic Design)
Backend API Service	http://localhost:5000 (Node.js + Express REST API + Background Cron)
Database Architecture	Dual-Engine: Zero-Config SQLite (Embedded) + Enterprise MySQL 8.0
Date of Generation	October 2026 | Technical Architecture Review

1. Executive Summary & Problem Context

Under prevailing AICTE statutory norms, thousands of engineering and technical institutions across India are mandatorily required to pledge substantial financial security deposits (typically ranging from ₹25 Lakhs to ₹1 Crore+ per institute) in the form of Fixed Deposit Receipts (FDRs) jointly held with AICTE. These deposits safeguard academic infrastructure, student interests, and faculty commitments.

Traditionally, the management, renewal, tracking, and encashment/release of these high-value instruments has suffered from acute operational bottlenecks:

- **Manual Verification & Paper Vulnerability:** Institutes submit paper FDR certificates by post or couriers, leading to physical wear, misplacement risks, and lengthy manual record-keeping in decentralized spreadsheets.
- **Maturity Oversight & Revenue Leakage:** Institutions and regulatory authorities regularly miss the 90-day/30-day renewal windows, causing FDRs to involuntarily enter low-interest auto-renewal or lapse into dormant accounts.
- **Opacity & Lack of Real-Time Analytics:** In manual workflows, regulatory officers lack a centralized real-time dashboard of total pledged security deposits, verification backlogs, or bank-wise distributions across states.
- **Absence of Compliance Audit Trails:** There was no cryptographically signed, immutable record of which officer approved an FDR, who requested a release, or when verification was completed.

> 📌 **THE FDRFLOW TRANSFORMATION:** FDRFlow completely eliminates these friction points by establishing an automated,

> unified, role-governed digital ecosystem where institutions submit FDRs electronically with proof documents, AICTE officers systematically verify instruments against banking standards, background daemons trigger proactive multi-tier maturity notifications, and all state transitions are immutably logged into an auditable ledger.

2. System Architecture & Tech Stack Matrix

FDRFlow follows an enterprise-grade, clean three-tier architecture with separation of concerns across the presentation layer, RESTful service layer, and persistence/daemon tier.

2.1 Architectural Components

- **Presentation Tier (Client):** Single Page Application (SPA) providing sub-millisecond route transitions, glassmorphic dark-mode dashboards, interactive lifecycle state machines, tabular filters, and dynamic forms with hot toasts.
- **API & Business Logic Tier:** Express.js REST application hosting JWT token authenticators, RBAC policy enforcement interceptors, Multer file upload storage, and controller business logic.
- **Data Persistence Tier:** Object-Relational Mapping (ORM) layer featuring a dual-dialect engine (instant zero-config SQLite for local evaluation and rapid evaluation, plus high-throughput MySQL for production deployments).
- **Asynchronous Alert Engine:** Embedded node-cron daemon running scheduled daily sweeps to calculate maturity thresholds (90, 30, and 7 days) and dispatch alerts.

2.2 Technology Stack Breakdown

Layer	Technology Selected	Architectural Justification & Role
Frontend Core	React 18 + Vite SPA	Lightning-fast HMR build, declarative component model, clean state management with AuthContext.
Styling System	Vanilla CSS Design Tokens	Zero-dependency CSS custom properties, responsive layout, glassmorphic cards, custom accessibility.
Iconography & UI	Lucide React + Hot Toast	Lightweight vector icons and non-blocking asynchronous floating feedback alerts.
Backend Runtime	Node.js (ES Modules) + Express	Asynchronous non-blocking event-driven execution, clean middleware pipeline, scalable REST APIs.
ORM & Modeling	Sequelize ORM v6	Model-driven schema definitions, automatic association joins, parameterized SQL injections defense.
Primary Database	SQLite (Embedded Zero-Config)	Delivers immediate plug-and-play local execution without installing external services.
Production DB	MySQL 8.0+	Supports enterprise scaling, transactions, foreign key indexing, and strict ACID guarantees.
Authentication	JWT + Bcrypt.js (10 Rounds)	Stateless bearer token exchange with salted cryptographic password hashing.

File Storage	Multer Multipart Engine	Handles authenticated FDR certificate scans and PDF bank guarantee proofs safely.
Background Cron	node-cron Scheduler	Automated daily checks evaluating FDR maturities and generating preemptive compliance alerts.

3. Features Implemented in the Live Prototype

The FDRFlow system implements the complete functional lifecycle envisioned by AICTE guidelines and the SIH Problem Statement:

3.1 Role-Based Access Control (RBAC) & Authentication

- **AICTE Officer (Regulatory Admin):** Complete authority to inspect all institutions, verify deposited FDRs, record official verification notes, advance workflow states, and authorize FDR renewals or fund releases.
- **Institution Admin (Depositor):** Dedicated view allowing institute authorized representatives (e.g. IIT Delhi, NIT Trichy, BITS Pilani) to register new FDR instruments, upload certificate copies, monitor status, and request timely renewals.
- **Compliance Auditor:** Read-only forensic view with full access to the complete, immutable system audit log, status transition timelines, and verification timestamps.
- **One-Click Quick Login Chips:** Login screen features interactive one-click role chips allowing evaluators to instantly switch between AICTE Officer, Auditor, and Institution Admin without manual credential typing.

3.2 Five-Stage FDR Lifecycle State Machine

FDRFlow enforces strict lifecycle state transitions, preventing unauthorized or out-of-order state mutations:

- **Stage 1: Submitted —** Institution submits the FDR instrument details, bank guarantee number, pledge amount, maturity date, and certificate attachment.
- **Stage 2: Under Verification —** AICTE Officer reviews banking details against bank confirmation letters, records verification notes, and marks instrument as verified or flags discrepancies.
- **Stage 3: Maturity Pending —** System tracks deposit during its active validity period; as the instrument approaches its deadline (90, 30, or 7 days), automated alerts flag pending action.
- **Stage 4: Renewal / Release Requested —** Institution requests deposit extension, or submits formal request for release/encashment upon completion of AICTE statutory pledge period.
- **Stage 5: Released / Expired —** Formal authorization by competent authority releasing funds back to the institution or closing the instrument upon completion.

3.3 Executive Dashboard & Real-Time KPI Analytics

- **Total Deposit Value Portfolio:** Aggregates total financial security value across all accredited technical universities (e.g. ₹3.88 Crores in demo registry).
- **Pending Verifications Counter:** Highlights fresh submissions awaiting regulatory officer review to prevent administrative backlogs.
- **Maturity Watchlist Widget:** Proactively isolates deposits reaching maturity within the next 30 to 90 days, preventing inadvertent auto-renewals.

- **Status Breakdown Bar:** Distribution breakdown visualizer displaying counts for Submitted, Under Verification, Maturity Pending, and Released statuses.

3.4 Interactive FDR Registry with Multi-Criteria Filtering

- **Status Pipeline Tabs:** Instantly filter deposits by 'All', 'Submitted', 'Under Verification', 'Maturity Pending', 'Renewal Requested', or 'Released'.
- **Multi-Column Instant Search:** Real-time search across Bank Name (SBI, PNB, BOB, HDFC, ICICI, etc.), Instrument Reference Number, and Institution Name.
- **Dynamic Status Badges:** Color-coded indicators with pulse animation for urgent attention states (e.g., Red for overdue/renewal requested, Amber for under verification, Green for verified).

3.5 Deep FDR Detail Inspection & Verification Workstation

- **Graphical Lifecycle Stepper:** Visual step indicator displaying current position in the 5-phase journey with completed, active, and upcoming indicators.
- **Financial Instrument Dossier:** Detailed breakdown of Institution, Bank & Branch, Instrument Ref, Principal Amount, Interest Rate, Issue Date, and Maturity Date.
- **Verification Action Form:** Exclusive to AICTE Officers — allows marking 'Verified', 'Discrepancy Found', or 'Manual Review' with mandatory evidence notes.
- **Workflow Transition Controller:** Allows authorized transition of state with mandatory rationale logged into audit records.
- **FDR-Specific Audit Trail:** Reverse chronological timeline showing every action, actor, timestamp, and notes associated with the specific FDR.

3.6 Automated Background Alert & Maturity Notification Scheduler

- **Proactive Monitoring Daemon:** An automated node-cron daemon checks the database daily at 08:00 AM.
- **Automated Multi-Tier Triggers:** Computes delta days = (Maturity Date - Current Date). Automatically categorizes deposits into 90-Day Warning, 30-Day Warning, 7-Day Critical, and Overdue.
- **Dual Notification Delivery:** Creates pending alert records and dispatches email notifications to both the Institution Admin and the AICTE nodal officer.

3.7 Immutable Audit Trail & Regulatory Compliance Logging

- **Universal Event Capture:** Every critical operation (User Login, FDR Submission, Verification Action, Status Change, Document Upload) is automatically recorded.
- **Granular Change Metadata:** Each audit entry stores FDR ID, Action Name, Previous Status, New Status, Performed By (User ID & Name), Timestamp, and Free-text rationale notes.
- **Dedicated Audit Explorer:** Dedicated searchable audit explorer with pagination, action-type color tags, and user attribution for oversight teams.

4. Novel Innovations & 'New Things' in FDRFlow

Beyond standard CRUD portals, FDRFlow introduces several high-impact software and architectural innovations designed specifically for regulatory efficiency and hackathon evaluation:

4.1 Smart Dual-Engine Database Abstraction (Zero-Config Portability)

A frequent pitfall in prototype deployments is the dependency on external relational database daemons (like MySQL or PostgreSQL) being installed and running on the evaluation machine. FDRFlow features a smart Sequelize connection layer that seamlessly operates in two modes:

- **Mode A: Embedded SQLite (Zero-Config Local Evaluation):** Out of the box, FDRFlow connects to an embedded SQLite database (`fdrflow.sqlite`). It requires zero installation, zero external services, and runs identically on Windows, Linux, or macOS.
- **Mode B: Enterprise MySQL:** By simply updating `.env` (`DB_DIALECT=mysql`), the identical codebase connects to an enterprise MySQL server with connection pooling and high concurrency.

4.2 Automated In-Memory Seeding Engine

Upon startup, FDRFlow automatically queries its user and institute tables. If empty, it executes an automated seeding sequence generating realistic, real-world data:

- **5 Accredited Institutions:** Pre-configures IIT Delhi, NIT Trichy, BITS Pilani, VIT Vellore, and Amity University.
- **5 Pre-configured Role Personas:** Configures AICTE Regulatory Officer, Compliance Auditor, and 3 Institution Administrators with pre-hashed Bcrypt passwords.
- **10 Diverse FDR Records:** Seeds 10 realistic multi-lakh and multi-crore FDRs distributed across State Bank of India, PNB, Bank of Baroda, Canara Bank, HDFC, ICICI, Union Bank, and Bank of India.
- **Initial Audit Trail:** Populates realistic historical status updates and verification logs to immediately demonstrate compliance reporting.

4.3 Evaluator-Centric One-Click Persona Switcher

To streamline live judging and presentations, the login portal features dedicated quick-login action buttons (`AICTE Officer`, `Auditor`, `IIT Delhi Admin`, `NIT Trichy Admin`). Evaluators do not need to memorize passwords or type credentials — a single click loads the role and navigates to the appropriate dashboard context.

4.4 Visual State Machine Lifecycle Stepper

Rather than displaying raw status text, FDRFlow features a 5-step interactive graphical stepper on every FDR detail view. Each phase dynamically reflects whether it has been completed, is currently active, or is awaiting future actions. This provides instant visual clarity for both university officials and government auditors.

4.5 Multi-Tier Automated Proactive Risk Detection

Instead of relying on human vigilance to identify expiring deposits, the proactive background cron scanner automatically calculates risk horizons (90 days, 30 days, 7 days) and triggers targeted alerts, eliminating the risk of accidental revenue lapses or forgotten bank guarantees.

5. Database Schema & Entity Relationships

The FDRFlow relational model consists of 7 normalized entities with strict foreign key constraints:

Entity Name	Primary Attributes	Business Logic & Relationships

institutions	id, name, address, contact_email, phone, approved_status	Stores accredited universities and colleges. Has many Users and FDRRecords.
users	id, name, email, password_hash, role, institution_id, is_active	System actors. Role is ENUM('aicte_officer', 'institution_admin', 'auditor'). Belongs to Institution.
fdr_records	id, institution_id, bank_name, instrument_ref, amount, dates, status	Core financial instrument record. Tracks bank, amounts, dates, and 7 lifecycle statuses.
fdr_documents	id, fdr_id, file_name, file_path, file_type, file_size, uploaded_by	Stores scanned copy of FDR certificates and bank verification confirmation letters.
audit_trail	id, fdr_id, action, old_status, new_status, performed_by, notes	Immutable compliance ledger capturing every state change, user login, and administrative action.
verifications	id, fdr_id, verified_by, verification_status, evidence_notes, verified_at	Formal regulatory verification records with status ENUM('verified', 'discrepancy_found', etc.).
alerts	id, fdr_id, alert_type, recipient_email, scheduled_at, sent_at, status	Scheduled and dispatched notifications for 90-day, 30-day, 7-day maturity warnings.

6. REST API Endpoints Catalogue

Method	Endpoint Path	Access Role	Functional Description
POST	/api/auth/login	Public	Authenticates user credentials, generates JWT token, and logs login action.
GET	/api/auth/me	Authenticated	Returns authenticated user profile, assigned role, and institution affiliation.
GET	/api/fdrs/stats	Authenticated	Calculates aggregated portfolio statistics (Total Value, Pending, Maturing, Counts).
GET	/api/fdrs	Authenticated	Returns filtered, searched, and paginated FDR records scoped by user role.
POST	/api/fdrs	Institution / Officer	Registers fresh FDR record and initiates verification stage.

GET	/api/fdrs/:id	Authenticated	Retrieves complete FDR instrument dossier, documents, and audit trail.
PUT	/api/fdrs/:id/status	Officer / Inst. Admin	Executes authorized state transition with mandatory rationale.
POST	/api/fdrs/:id/verify	AICTE Officer Only	Submits formal officer verification findings (verified, discrepancy, etc.).
POST	/api/fdrs/:id/documents	Authenticated	Uploads scanned FDR document proof using Multer multipart handling.
GET	/api/fdrs/audit/trail	Authenticated	Returns paginated immutable system audit log with actor details.
GET	/api/institutions	Officer / Auditor	Retrieves comprehensive list of technical institutions and their deposit counts.
GET	/api/health	Public	System health diagnostic endpoint returning server status and timestamp.

7. Deployment & Verification Guide

7.1 Prerequisites

- **Runtime:** Node.js v18.0+ or higher and npm package manager.
- **Web Browser:** Any standard web browser (Google Chrome, Microsoft Edge, Mozilla Firefox).
- **Database Server:** Optional (Embedded SQLite is pre-configured and active by default).

7.2 How to Start the System Locally

To run the application from the repository root:

- **Step 1 (Start Backend):** Open PowerShell or Terminal and execute:

```
cd fdrflow\server
npm start
```

- **Step 2 (Start Frontend):** Open a second terminal window and execute:

```
cd fdrflow\client
npm run dev
```

- **Step 3 (Access Web App):** Open your browser and navigate to http://localhost:5173

7.3 Recommended Evaluator Walkthrough

- **1. Officer Overview:** Click the 'AICTE Officer' quick button on the login screen and click 'Sign In'.
- **2. Inspect KPIs:** Observe the Total Portfolio Value (₹3.88 Crores), 4 Pending Verifications, and 2 Maturing Soon warnings.

- **3. Workflow Stepper:** Click on 'State Bank of India (SBI/FDR/2024/001)' to open the detail view. Inspect the visual lifecycle stepper.
- **4. Complete Verification:** As an AICTE Officer, submit a verification decision (e.g. 'Verified - Joint discharge signed') and witness the audit trail update immediately.
- **5. Audit Compliance:** Click 'Audit Trail' in the sidebar to review the chronological, immutable log of all historical system operations.
- **6. Test Multi-Tenancy:** Logout and click 'IIT Delhi Admin'. Notice that the institution view is strictly partitioned to only show deposits belonging to IIT Delhi.